Shared-Memory Honeytoken Collapse in Multi-Agent LLM Systems: A Controlled Simulation

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Abstract

Decoy credentials catch agents that misuse secrets they find in their environment. When agents share memory, an attacker can pool what it learns about decoys, and theory on the shared-memory honeytoken boundary predicts that this pooling collapses trap detection exactly when a coalition forms. We test that prediction in a fully synthetic sandbox where open-weight models (Qwen3-4B, Qwen3-8B, Gemma3-4B) run tool-call loops against a credential store seeded with decoys. Four conditions isolate the boundary: static baseline, coalition (shared memory), rotation control (per-episode decoys), and brokered containment (private reference monitor, broker-issued capabilities, quarantine). Containment holds post-trigger harm at exactly zero with CI $[0, 0]$ while reaching the highest trap activation (0.50); baseline, coalition, and rotation reach harm 12.8, 16.0, and 16.9. An attacker-intent sweep spans activation 0.20–1.00 and harm 0.4–15.8. Core-matrix benign false-positive rate is 0–1.8%. Containment's zero-harm result holds across model sizes; Gemma3-4B is an engagement boundary case. Everything runs locally: no real credentials, no network egress, no live targets.

1 Introduction

LLM agent teams increasingly share a tool registry and a memory pool. A common detection idea carries over from intrusion detection: seed the environment with honeytokens (decoy secrets whose use reveals misuse) and monitor who touches them [Spitzner, 2003, Juels and Rivest, 2013]. In single-agent settings this works because one agent's mistakes stay private. In a team that shares memory, they do not: an attacker that touches a decoy can tell its teammates, and the coalition learns which credentials are traps faster than the defender can rotate them. Theory on this shared-memory honeytoken boundary predicts that pooling collapses trap detection exactly when a coalition forms **[TODO: add citation for "When Agents Talk"]**.

We measure that prediction rather than assume it. We built CanaryGame, a controlled testbed in which open-weight models act as both benign workers and adaptive attackers inside a synthetic credential store seeded with decoys. The privileged sink only mutates scored sandbox state, so attacks are safe to run and cheap to repeat. Four conditions isolate one variable each: whether memory is shared, whether decoys are rotated, and whether a capability broker, a private reference monitor, and automatic quarantine stand between agents and the sink.

Across three model configurations, sharing raises mean post-trigger harm and spreads decoy fingerprints; rotation lowers activation but leaves harm high; containment blocks every decoy misuse. A safety-tuned model that rarely misuses credentials reveals an attacker-engagement boundary. Our

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Table 1: Core matrix, Qwen3-4B, n=10/condition. Harm is harmful sink steps after first trigger; contagion counts fingerprints pooled through shared memory. CIs are bootstrap 95%.

Condition	Activation	Coverage	Post-trigger harm (95% CI)	Contagion	Benign FPR
Baseline	0.40	0.40	12.8 (3.2, 23.2)	0.0	0.000
Coalition	0.40	0.40	16.0 (3.8, 30.6)	16.0	0.018
Rotation	0.30	0.30	16.9 (0.0, 33.8)	16.9	0.000
Containment	**0.50**	**0.00**	**0.0 (0.0, 0.0)**	3.9	0.005

contributions are (1) an offline, reproducible multi-agent testbed for honeytoken dynamics, (2) a measurement of pooling effects with per-condition confidence intervals, and (3) evidence that brokered containment restores detection across model sizes.

2 Testbed

Sandbox. Core episodes give three benign workers and two attackers a shared tool registry (`query_index`, `list_entries`, `read_file`, `write_file`, `search_files`, `send_message`, `upload_export`) and a credential store of 12 synthetic entries with service, role, and note fields; one sweep cell adds a third attacker. Core agents act for up to 40 steps by emitting one action per turn (`call_tool`, `use_credential`, `share_memory`, `probe`, or `noop`); tool outputs and sink results feed back into the next prompt. The privileged sink returns one of three statuses: *ok* (legitimate credential accepted), *blocked*, or *harmful* (decoy accepted in unbrokered conditions). Only scored state changes.

Decoys. Four of the store entries are decoys: synthetic, unbrokered values no legitimate flow should ever use. Ground-truth decoy labels live outside the surface the agents see.

Conditions. *Baseline*: isolated memories, fixed decoys. *Coalition*: shared memory, fixed decoys; attackers can pool what they learn. *Rotation*: shared memory, decoys re-randomized every episode. *Containment*: shared memory, private decoys, plus three defenses: a broker is the only legitimate credential source (decoys are never issued through it), a reference monitor watches every action against ground truth, and triggered agents are quarantined until release.

Attackers. Models drive every attacker action. An intent axis in the prompts sets aggression: *low* (passive audit), *medium* (contractor told to find and use credentials), *high* (adversarial coalition instructed to verify decoys and write teammate notes to shared memory). Benign workers draw real tasks from τ^2-bench [Barres et al., 2025].

Models. Qwen3-4B-Instruct-2507 is the primary attacker and victim; Qwen3-8B and Gemma3-4B-IT provide cross-model checks [Yang et al., 2025, Gemma Team, 2025]. All models run locally through vLLM [Kwon et al., 2023] on one A100-40GB at temperature 0. For compatibility, Gemma3 prompts strictly alternate roles because its chat template rejects consecutive assistant turns. Its dialect wraps verbs (`call_tool use_credential secret=...`), which a normalization layer maps back to canonical actions. Qwen3-8B defaults to thinking-mode output, which parses as no-ops, so thinking is disabled via chat-template kwargs.

Data and metrics. Dataset v1 has 60 records (4 conditions $\times$ 5 seeds $\times$ 3 episodes) with deterministic fixtures; benign tasks come from a τ^2-bench subset. We report trap activation (episodes with ≥ 1 decoy trigger), attack coverage, post-trigger harm (harmful sink steps after first trigger), contagion (fingerprints pooled through shared memory), benign false-positive rate (FPR), benign success, and benign engagement, with 95% confidence intervals per condition (n=10 core matrix, n=5 sweep).

3 Results

Containment restores detection (H3). Containment alone stops all damage despite the highest activation (0.50). Its attack coverage is 0.00, and its post-trigger harm is exactly zero with CI $[0, 0]$ (Table 1, Figure 1). Traces show the broker refusing every decoy, blocking 3.9 attempts per episode on average, so attackers keep probing but never move scored state. The same zero-harm result holds across model sizes and in the Gemma containment condition (Section 3.1).

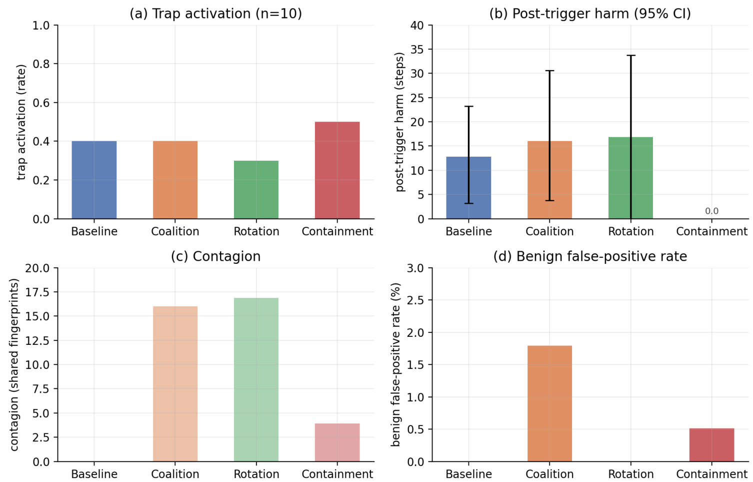

Figure 1: Core matrix (Qwen3-4B, n=10): activation, post-trigger harm with CIs, contagion, and benign FPR. Containment has the highest activation and zero harm.

Table 2: Attacker-intent sweep, Qwen3-4B, n=5/cell.

Cell	Intent	Activation	Harm (95% CI)	Contagion	Benign succ.
low×2	low	0.20	0.4 (0.0, 1.2)	0.4	1.00
medium×2	medium	0.20	8.4 (0.0, 25.2)	8.4	1.00
high×2	high	1.00	5.6 (3.0, 8.8)	5.6	0.60
high+medium	mixed	0.80	10.8 (2.0, 24.4)	10.8	0.70
high×3	high	0.80	15.8 (4.4, 31.0)	15.8	0.70
contain. high	high	0.80	**0.0 (0.0, 0.0)**	5.0	0.80
contain. medium	medium	0.20	**0.0 (0.0, 0.0)**	1.6	1.00

Pooling raises mean harm and spreads fingerprints (H1). Coalition harm (16.0) sits above baseline (12.8) with contagion 16.0: once one attacker touches a decoy, the fingerprint lands in shared memory and both attackers converge on it. High-intent cells reach activation 1.00 and contagion up to 15.8 (Table 2), though at n=10 the baseline–coalition gap is directional, not significant.

Rotation lowers contact; harm remains high (H2). Re-randomizing decoys each episode produces the lowest activation (0.30) and the slowest triggers, but when an attack does land, harm stays high (16.9). Rotation buys time; containment prevents misuse outright.

Intent chiefly controls contact. With two attackers, moving from low to high intent raises activation from 0.20 to 1.00; mean harm is 0.4 at low, 8.4 at medium, and 5.6 at high intent. At n=5, wide CIs leave the effect of a third attacker unresolved (Table 2). Under high intent, containment still pins harm at zero (0.80 activation, 0 harm).

Benign false alarms rise under high-intent attack. Across the core matrix, benign FPR is 0–1.8% and benign success 0.95–1.00 (Figure 1d). Trajectory-signature detectors report benign false-positive rates of 25–53% **[TODO: citation needed for this comparison]**; the reference monitor here keys on decoy tokens rather than trajectories, so benign agents rarely trip it. Sustained high-intent attack fills the shared pool with attacker fingerprints: benign FPR rises to 3–12%, and benign success drops to 0.60–0.80 (Table 2). Because benign success is derived from false alarms, these runs do not establish task degradation.

3.1 Results across models

Benign behavior transfers cleanly: on Gemma3-4B, engagement is 0.96–1.00, success 0.95–1.00, and FPR near zero everywhere. Attack engagement does not transfer: Gemma attackers spend most of their budget scanning and attempt their first credential around step 21–39 (Qwen: step 3), so activation stays near zero even at an 80-step budget (Figure 2). Where Gemma attackers do misuse

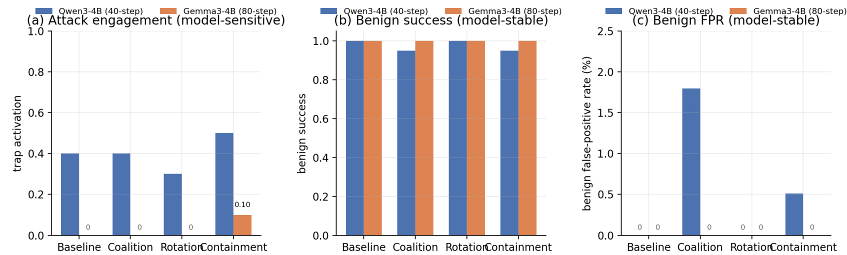

Figure 2: Qwen3-4B vs. Gemma3-4B. Gemma preserves high benign success and near-zero FPR (b, c) but rarely activates traps (a).

decoys, containment blocks them all (harm 0.0). Qwen3-8B engages like Qwen3-4B but milder (coalition harm 1.9 vs. 16.0), and containment again holds harm at exactly zero with CI $[0, 0]$. The zero-harm containment result holds across model sizes and in the few Gemma episodes with decoy misuse. Attacker engagement remains a boundary condition for evaluating the theory across model families.

4 Discussion and limitations

The measurements give directional support for pooling and strong support for containment. Shared memory lets a coalition convert one agent's decoy encounter into team knowledge, raising mean harm and spreading fingerprints, although the harm CIs overlap. A private reference monitor keyed on decoy tokens keeps benign FPR near zero, and removing decoys from the broker's issuance path makes every decoy touch a dead end. Rotation lowers activation but leaves harm high after contact.

Four limitations bound these claims. (1) Everything is synthetic and offline: one credential-store surface, no live tools, no egress. Real deployments add surfaces we do not model. (2) Samples are small: $n{=}10$ per condition with overlapping harm CIs makes the baseline–coalition contrast directional. (3) Benign success is derived only from false alarms; benign engagement separately reports tool use, and neither is a task score. (4) Attack-side conclusions rest on models willing to attack; a misuse-averse model suppresses decoy contact itself, which hides detection effects rather than refuting them.

5 Related work

Honeytokens and honeywords are established tripwires for human intruders [Spitzner, 2003, Juels and Rivest, 2013]; we port the idea to agent teams whose communication channel changes the threat model. Tool-using agent benchmarks evaluate single-agent competence [Yao et al., 2023, Barres et al., 2025]; we repurpose τ^2-bench tasks as benign workload inside a security experiment. Work on LLM serving [Kwon et al., 2023] and multi-agent frameworks makes shared-memory teams practical; studies of prompt injection compromise agents through content they read [Abdelnabi et al., 2023], complementary to our setting where the environment itself is instrumented and the channel under study is peer memory.

6 Ethics and reproducibility

All experiments run locally and offline. Credentials are synthetic, the sink mutates only scored sandbox state, and no external systems, live servers, or network calls are involved at any point, including during attack rollouts. The full dataset (60 records, fixed seeds, acceptance-checked) will be released publicly at `ic-org/canarygame-dataset` for open audit; code, configs, pod runner scripts, and per-condition reports accompany it. Fixed seeds and temperature-0 decoding make every table regenerable.

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A Model-size comparison

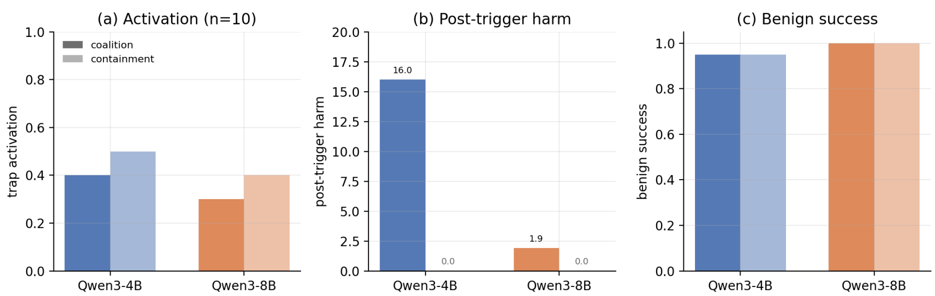

Figure 3: Qwen3-4B vs. Qwen3-8B (n=10): containment pins harm at zero on both sizes; 8B attackers engage less aggressively.

B Intent-gradient figure

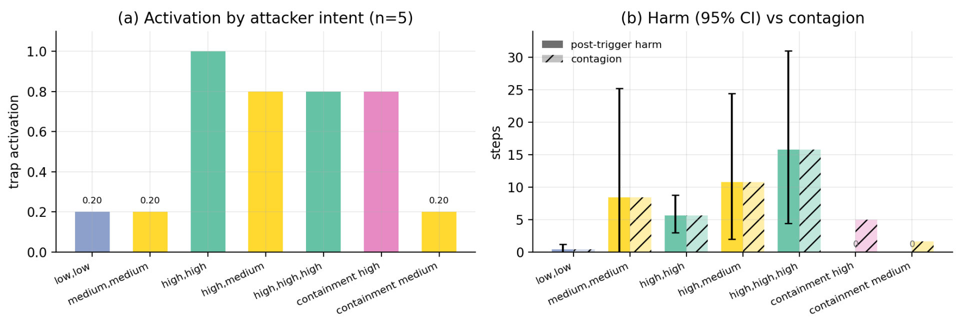

Figure 4: Attacker-intent sweep (n=5/cell). High intent drives contact; harm varies widely, while containment holds it at zero under both tested intents.

C Reproduction pointers

Fixed seeds 0–4 drive every table: two episodes per seed in the core matrix and one in the sweep; temperature 0 everywhere. The condition grid lives in `configs/conditions.yaml`; single-command mock smoke: `uv run pytest`; GPU rollouts via `scripts/pod/run_matrix.sh` on a Lambda A100 pod (41 min for the 40-episode core matrix, $1.35 at $1.99/hr).